

PROFESSIONAL SUMMARY

Biostatistician developing statistical methods for reliable clinical evidence from selectively observed data. My work bridges EHR-derived real-world evidence and early-phase drug development, with methodological contributions in causal inference, survival and longitudinal data analysis, covariate adjustment, and Bayesian dose optimization. My published first-author methodology papers appear in *Biometrika*, *The Annals of Applied Statistics*, and *Statistics in Medicine*, with related manuscripts under revision at *Journal of the American Statistical Association* and *Biometrics*. My current EHR research spans All of Us, UK Biobank, Yale New Haven Health System, the Michigan Genomics Initiative, and Epic Cosmos; includes the CRAN package CIMEHR; and is reflected in a submitted NIH R01 application currently under review, for which I serve as Multiple Principal Investigator with Bhramar Mukherjee.

RESEARCH EXPERIENCE

Yale University

Associate Research Scientist (Mentor: Dr. Bhramar Mukherjee)

New Haven, CT

Aug 2025 –Present

- Developing joint modeling approaches for longitudinal EHR data with informative visit and observation processes.
- Studying testable implications of missing-at-random assumptions in clinical recording processes.

The University of Texas MD Anderson Cancer Center

Graduate Research Assistant (Mentor: Dr. Ruitao Lin)

Houston, TX

Aug 2021 –Aug 2025

- Developed statistical methodology for early-phase oncology trials, including Bayesian dose-finding designs, dose optimization, and dose-response estimation.

EDUCATION

University of Texas Health Science Center

Ph.D. in Biostatistics and Data Science, Advisor: Ruitao Lin

Houston, TX

2021–2025

- Thesis: “Methodological Advances in Early Phase Clinical Trial Design and Analysis.”
- Graduate Research Assistant at The University of Texas MD Anderson Cancer Center.

National Tsing Hua University

M.S. in Statistics, Advisor: Yu-Jen Cheng

Hsinchu, Taiwan

2015–2017

- Thesis: “Regularized Interaction Boosting with Right Censored Data.”

PUBLICATIONS

Methodology Papers

- [1] S McGrath, **CH Yang**, J Kimmelman, O Ozturk, R Steele, and A Benedetti. “Meta-Analysis of Median Survival Times With Inverse-Variance Weighting”. In: *Statistics in Medicine* 45.8-9 (2026), e70533.
- [2] **CH Yang**, PF Thall, D Marin, SY Belay, and R Lin. “BAR12: Bayesian Autoregressive Phase 1-2 Design for Cell Therapy Trials With Manufacturing Changes”. In: *Statistics in Medicine* 45.8-9 (2026), e70551.

- [3] **CH Yang**, PF Thall, and R Lin. “Dose exploration, monitoring, and optimization using biological and clinical outcomes”. In: *The Annals of Applied Statistics* 19.4 (2025), pp. 2599–2617.
- [4] **CH Yang**, G Cheng, and R Lin. “On the relative conservativeness of Bayesian logistic regression method in oncology dose-finding studies”. In: *Pharmaceutical statistics* 23.4 (2024), pp. 585–594.
- [5] **CH Yang** and YJ Cheng. “A model-free variable screening method for optimal treatment regimes with high-dimensional survival data”. In: *Biometrika* 111.4 (2024), pp. 1369–1386.
- [6] **CH Yang**, E Kwiatkowski, JJ Lee, and R Lin. “REDOMA: Bayesian random-effects dose-optimization meta-analysis using spike-and-slab priors”. In: *Statistics in medicine* 43.18 (2024), pp. 3484–3502.

Collaborative Papers

- [7] DN Yeboa, BT Whitfield, R Lin, CL Ejezie, TA Swanson, TH Beckham, C Wang, B De, S Perni, MC Tom, J Li, SL McGovern, R Harrison, NK Majd, VK Puduvalli, AE Aaroe, M Loghin, BJ O’Brien, AD Patel, CB Patel, JS Wefel, C Altintas Taslicay, M Gule-Monroe, AC Paulino, MF McAleer, DR Grosshans, AJ Ghia, W Jiang, C Chung, M Maor, **CH Yang**, MA Gubbiotti, C Kamiya-Matsuoka, LY Ballester, SP Weathers, and JT Huse. “Prospective phase II clinical trial of molecular glioblastoma (historical grade 2 and 3 IDH wildtype gliomas) preliminary novel exploratory analyses”. In: *Journal of Neuro-Oncology* 176.1 (2026), pp. 1–11.
- [8] KM Moody, MC Swartz, Z Gresham, M Askins, L Cahalan, C Geistkemper, A Heaton, R Lin, M Melo, A Rajan, M Smith, P Tewari, K Williams, **CH Yang**, and R Robert. “A Virtual Wellness Program to Enhance Well-being for Pediatric Oncology Staff during the COVID-19 Pandemic”. In: *Adv Cancer Educ Qual Improv* 1.1 (2025), a18. doi: 10.52519/ACEQI.25.1.1.a18.

Preprints / Under Review

*Equal contribution

- [9] X Chen, **CH Yang**, Z Wu, and B Mukherjee. *Prediction-based Inference in Electronic Health Record (EHR)-linked Biobanks*. Submitted to *Statistics in Medicine*. 2026. arXiv: 2603.14356.
- [10] J Du *, **CH Yang** *, X Shi, and B Mukherjee. “Analyzing Longitudinal Outcomes from Electronic Health Records: A Joint Modeling Approach to Handle Informative Visits and Observations”. Revised and resubmitted to *Journal of the American Statistical Association*. 2026.
- [11] **CH Yang**, X Shi, and B Mukherjee. *Joint Modeling of Longitudinal EHR Data with Shared Random Effects for Informative Visiting and Observation Processes*. Revised and resubmitted to *Biometrics*. 2026. arXiv: 2602.15374.
- [12] Y Hou, T Ward, **CH Yang**, E Jernigan, G Caturegli, D Boffa, and B Mukherjee. *Assessing Statistical Practices of Existing Artificial Intelligence (AI) Models for Lung Cancer Detection, Prognosis, and Risk Prediction: A Cross-Sectional Meta-Research Study Supplemented by Human and Large Language Model (LLM)-Directed Quality Appraisal*. 2025. doi: 10.64898/2025.12.23.25342849v1. medRxiv: 2025.12.23.25342849.

WORKING PAPERS

- [1] **CH Yang** and R Lin. “MCP-ModX: Multiple Comparison Procedure and Modeling with Model-Assisted Covariate Adjustment under Missing Outcomes”. In preparation. 2026.
- [2] **CH Yang**, S McGrath, RD Shah, X Shi, and B Mukherjee. “A Test for Missing at Random in Longitudinal Biomarkers from Electronic Health Records Using Visits Without Measurement”. In preparation. 2026.

- [3] **CH Yang**, M Salvatore, H Lu, Z Zhu, P Tennant, X Shi, F Li, and B Mukherjee. “Who seeks care, and what gets measured? Understanding the distinct mechanisms behind visit and observation processes in multi-center electronic health records”. In preparation. 2026.

SOFTWARE

- [1] **CH Yang** and Y Hou. *CIMEHR: Gaussian Clinically Informative Visiting and Observation Processes in Electronic Health Record Data*. R package version 0.1.0. 2026.

SUBMITTED GRANTS

NIH R01

Developing a Statistical Toolkit for Modeling Clinically Informative Reporting in Electronic Health Records. Under review

- Role: Multiple Principal Investigator with Bhramar Mukherjee
- Proposed project period: May 2027–Apr 2032

PRESENTATIONS

Invited Seminars

- **Yang CH**. “A Joint Modeling Approach for Longitudinal EHR Data with Informative Visiting and Observation Processes.” Yale Public Health Data Science and Data Equity Seminar, New Haven, CT, Jan 2026.
- **Yang CH**. “A Joint Modeling Approach for Longitudinal EHR Data with Informative Visiting and Observation Processes.” Penn Medicine, University of Pennsylvania, Philadelphia, PA, Mar 2026.

Oral Presentations

- **Yang CH**. “Joint Modeling of Longitudinal EHR Data with Shared Random Effects for Informative Visiting and Observation Processes.” Joint Statistical Meetings, Boston, MA, Aug 2026. Oral presentation.

Poster Presentations

- **Yang CH**, Thall PF, Lin R. “Dose-Finding Design Guided by Early Biological Mediators and Clinical Outcomes.” ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Rockville, MD, Sep 2025.
- **Yang CH**, Thall PF, Lin R. “Mediator-Supported Design for Exploring and Optimizing Doses in Phase I-II Trials.” Statistics in Pharmaceuticals Conference, Storrs, CT, Aug 2025.

ADVISING AND MENTORING

- **Waveley Qiu**, PhD student in Biostatistics, YSPH; Research Co-Mentor with Bhramar Mukherjee 2026 –Present
- **Henan Xu**, Postdoctoral Associate, YSPH; Research Co-Mentor with Bhramar Mukherjee 2026 –Present

AWARDS AND HONORS

- **Elected Full Member**, Sigma Xi, The Scientific Research Honor Society 2025
- **Inducted Member**, Delta Omega Honorary Society in Public Health 2025

- **Student Travel Grant**, ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop 2025
- **Scholarship Award**, North America Taiwanese Engineers' Association (NATEA) 2023
- **Robert H. Bigelow Endowed Scholarship**, UT Health Science Center at Houston 2022, 2024
- **Academic Excellence Award**, National Tsing Hua University 2016, 2017
- **Merit Scholarship**, National Tsing Hua University 2015

TEACHING AND MENTORING EXPERIENCE

Big Data Summer Immersion at Yale (BDSY)

New Haven, CT

EHR Data Analysis Group Lead

Summer 2026

- Led undergraduate trainees in a mentored research project on electronic health record data analysis.
- Guided project design, EHR data structure, cohort definition, missingness mechanisms, and final presentation.

National Tsing Hua University

Hsinchu, Taiwan

Graduate Teaching Assistant

2017–2021

- STAT661000 Survival Analysis (Spring 2020, Spring 2021)
- STAT519100 Applied Multivariate Analysis (Spring 2017, Spring 2019)
- STAT541000 Linear Model (Fall 2018)

National Tsing Hua University

Hsinchu, Taiwan

College Teaching Assistant

2016–2020

- ECON303300 Econometrics (Fall 2019)
- STAT202000 Statistics (Fall 2016, Spring 2020)

PROFESSIONAL SERVICE

- **Session Chair**, “Statistical Research in Precision Medicine,” Joint Statistical Meetings, Boston, MA Aug 2026
- **Reviewer:**
Statistics in Medicine (2), *Nature* (1), *BMC Medical Research Methodology* (2), *Pharmaceutical Statistics* (1), *Journal of biopharmaceutical statistics* (1)